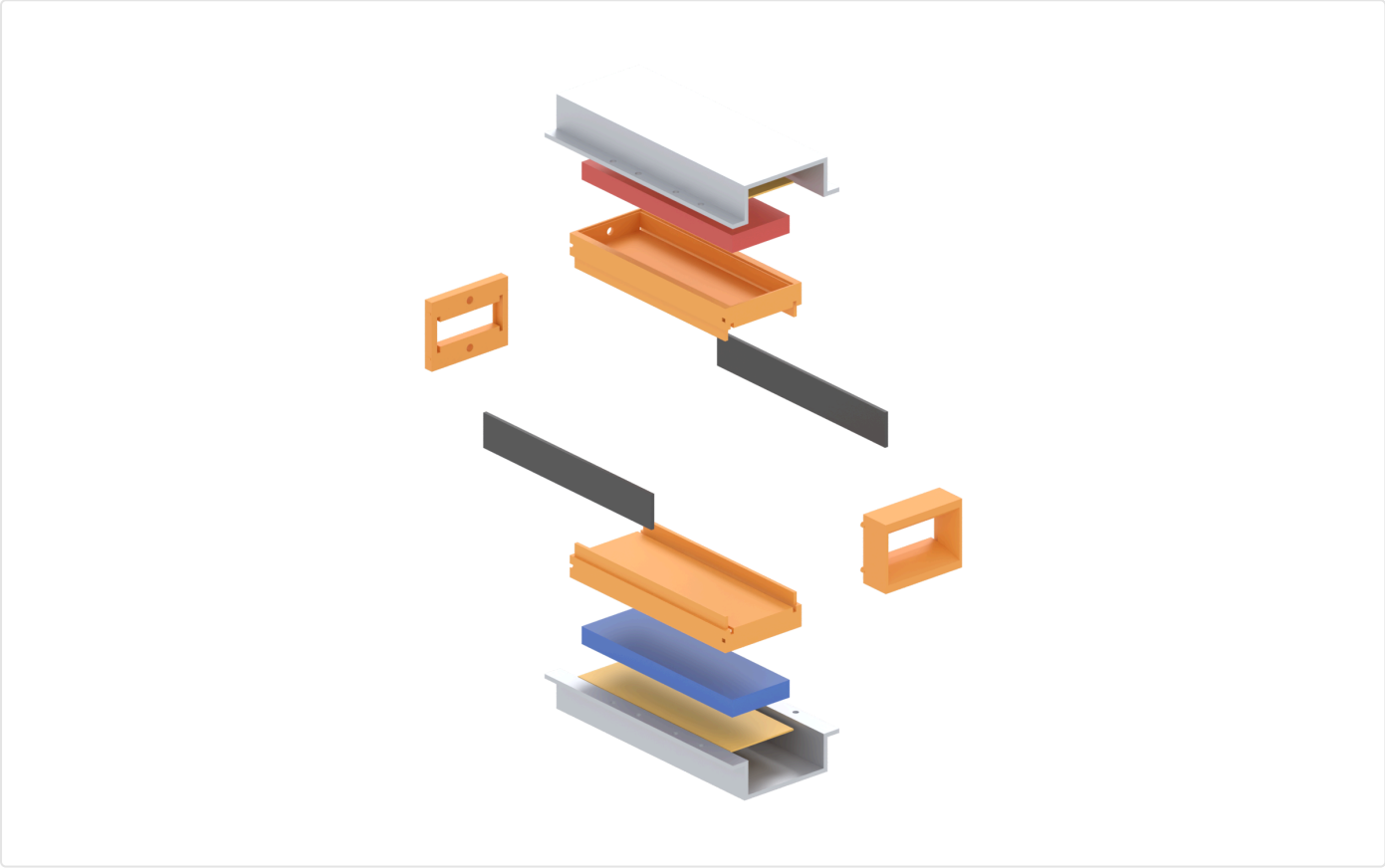
Magnetide v21-r2e — Manufacturing & Assembly

Reynard Filbert Tang · Kelas 10.8, Kelompok 1 · MHD seawater pump
Envelope 150 × 72 × 44 mm · bore 40 × 14 × 124 mm · all dimensions in mm
Every figure is a render of the delivered solid, not a drawing of it.

What you are building

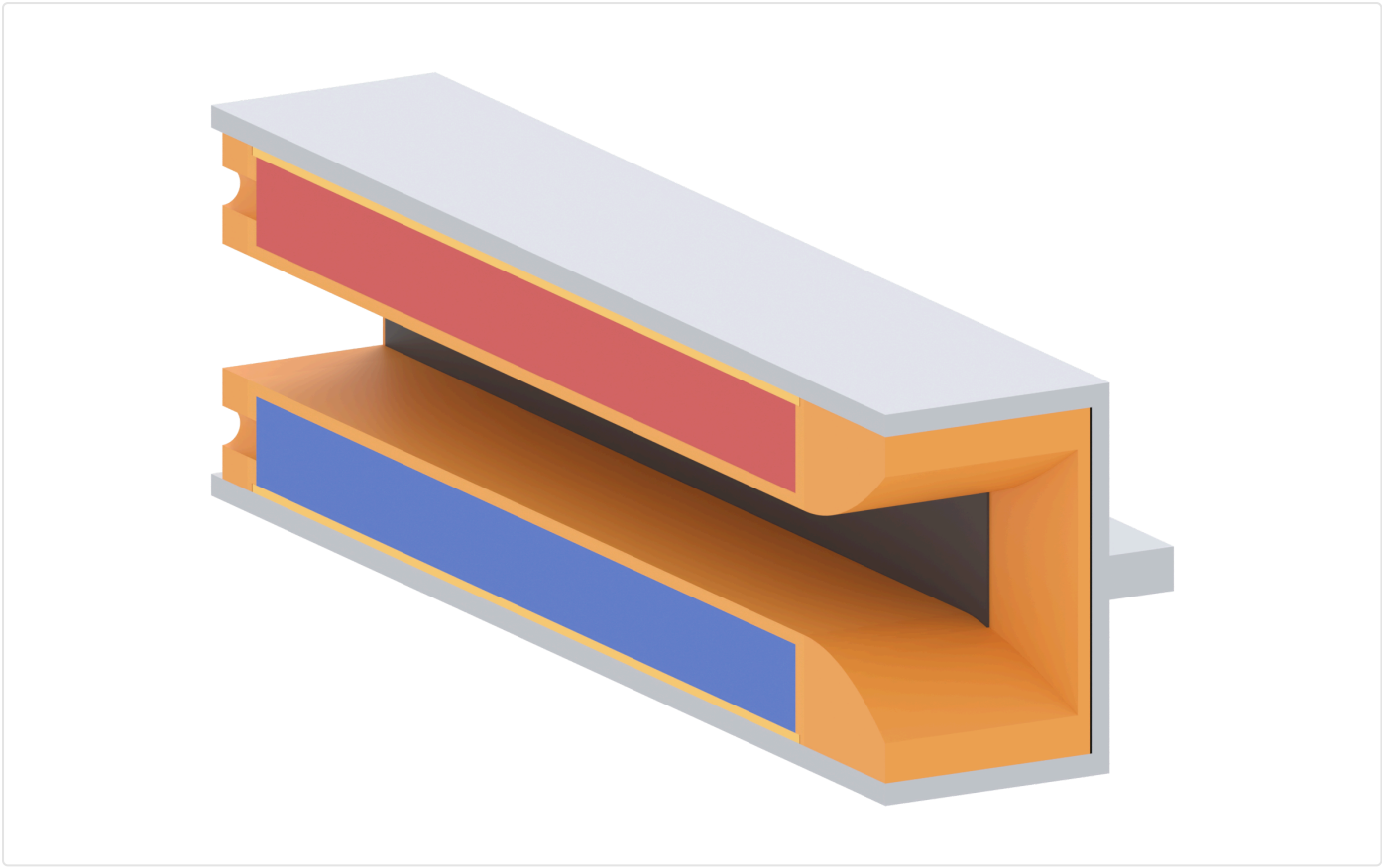
Brine runs through a 40 × 14 mm bore. A 483 mT field crosses it, current flows between two graphite plates on the side walls, and the Lorentz force pushes the water out the straight end. The steel is a folded 2.5 mm clamshell and it is the magnetic return path — not a cosmetic box.



Exploded, upper half. Bottom to top: steel shell, tray with its magnet cavity, four magnet bars, 1 mm cap, and the mating steel half. Bellmouth at the inlet (left), rear block with the lead exit at the outlet (right).

Stack, per half (z from the mid-plane)

z	Layer	Thickness	Function
0 – 7	bore	14 tall	water channel, 40 wide
7 – 8.5	tray floor	1.5	the only thing between water and magnets
8.5 – 18.5	magnet in cavity	10	walled in on all four sides
18.5 – 19.5	cap	1.0	locks the magnets down, flush with the steel
19.5 – 22	steel plate	2.5	flux return



Same model cut on the centre plane. You are looking at the bore, the 1.5 mm tray floor above it, the magnet sitting in the cavity, and the 1 mm cap between the magnet and the steel. The red and blue faces are the two magnet polarities.

Parts

Metal — `metal/`

File	Qty	Material	Notes
`clamshell-flat-2.5mm.dxf`	2	2.5 mm mild steel (DC01 / SPCC / AISI 1008)	Blank **107.83 × 150**. Laser cut, 8 × Ø3.4 holes, then folded twice. One blank per half; mirror one after folding.

Printed — `plastic/`

File	Qty	Size	Notes
`tray.stl`	2	124 × 51.5 × 19.5	Magnet cavity 120 × 40 × 10 with a 1 mm ledge for the cap. Print the floor side down.
`cap.stl`	2	121.7 × 42.7 × 1.0	Thin lid. Print flat, 100 % infill — it is a clamp, not a shell.
`bellmouth.stl`	1	20 × 52 × 39	48 × 30 mouth closing to 40 × 14. Print mouth-up, supports only in the bore.
`rear_block.stl`	1	5.5 × 52 × 39	Outlet. Has the 4.4 mm lead exit; keep that hole clear.

Bought

Item	Qty	Notes
N52 tile 20 × 10 × 5, nickel plated	96	48 per face: 6 along × 4 across × 2 deep. Two tiles stacked = one 10 mm bar.

Graphite plate, 134 × 14 × 2	2	Anode and cathode. 3 mm of each end is buried in plastic — 124 mm is wetted.
M3 × 16 bolt + nut + washer	8	Through the folded flanges at the split line.
M3 grub screw or 3 mm dowel	2	Optional registration pin — see step 6.
Epoxy (slow, 30 min+)	—	Magnets to tray. **Not** on the flange faces.

Manufacturing

Steel: laser + 2 folds

Parameter	Value	Where it comes from
Sheet thickness	2.5	one gauge for the whole half — walls and plate
Flat blank	107.83 × 150	56 plate + 2 × (19.96 wall + 5.96 flange)
Inside bend radius	0.2	sharp; a knife-edge die is fine at this gauge
K-factor	0.44	typical for mild steel at R/t ≈ 0.08
Bend allowance	2.042	$(R + K{\cdot}t){\cdot}\pi/2$ per 90° bend
Fold line, from the blank edge	47.96	marked on the DXF FOLD layer
Mass per half	317 g	7.85 g/cm³

> **Cut one test blank first.** K = 0.44 is nominal. If your brake gives K = 0.40 the flange shortens by 0.3 mm and the bolt holes drift — measure the folded part, not the drawing, before you cut the second blank.

> **The fold must be 90° up on both edges, and both flanges must land on the mid-plane.** Those two flange faces are the magnetic return path: they are bolted bare, steel on steel, no paint and no gasket. 1.5 mm of anything in that joint is 1.5 mm of air gap in series with the whole field.

Printing

Part	Layer	Walls	Infill	Orientation
tray	0.2	4	40 %	floor down; the cavity prints open-side up with no supports
cap	0.16	3	100 %	flat — it is 1 mm, any warp is a leak path
bellmouth	0.2	3	25 %	mouth up, bore vertical
rear_block	0.2	4	40 %	flat face down

PLA is fine for a demo run; PETG if the water is warm or the pump runs long. The tray floor is a 1.5 mm water barrier — **if it leaks, the magnets are gone**, so keep that face solid and inspect it against a light before assembly.

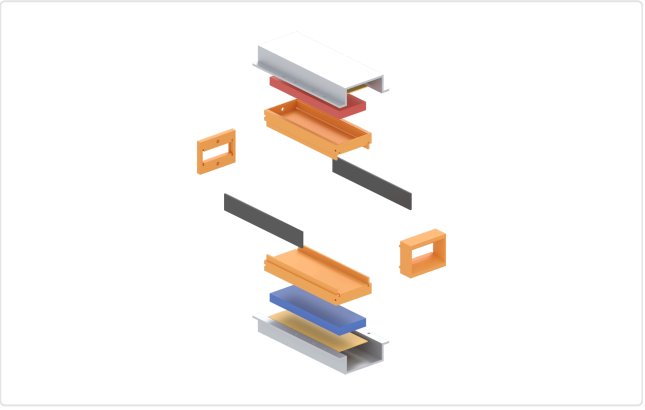
Build sequence

1 Fold the two steel halves

Cut two blanks from `clamshell-flat-2.5mm.dxf`. Fold 90° along both dashed lines: fold 1 at 47.96 from the edge (wall), fold 2 at 5.96 further in (flange).

Check the folded part: 56 mm across the plate, 22 mm from the plate to the flange face, holes on the split line.

2 Stack the magnets into the tray cavity



The tray is a cup, not a plate: floor on the bottom, walls on all four sides. Eight tiles per row, four rows, two layers deep — 48 per face. Two layers of 5 mm tiles = one 10 mm bar, filling the cavity 8.5 → 18.5.

Compass-check every row before the epoxy sets. All four rows on one face point the same way; the opposite face points the other way. If one row is flipped the pump stalls in that quarter of the bore.

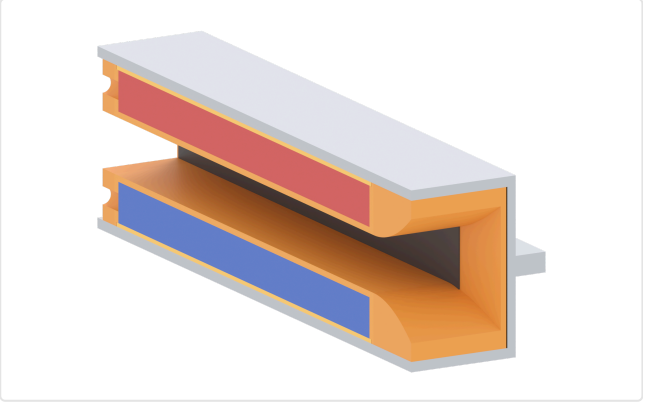
> Tiles are modelled as 9.7 mm wide bars with 0.3 mm clearance so they drop in. Do not press-fit them — a cracked tile is a dead tile and the nickel will corrode from the crack.

3 Cap it

Drop the 1 mm cap into the ledge in the tray rim. It sits on the magnets, its top face flush with the steel inner face. It is what stops the stack lifting when the field pulls.

No glue needed if the fit is right; a bead around the rim is acceptable, a puddle is not — excess epoxy that runs into the bore is a flow obstruction you cannot reach.

4 Seat the electrodes



Each 134 × 14 × 2 graphite plate slides into the 2.2 mm slot in the tray wall from the end. 3 mm of its height is buried in the tray, 14 mm is exposed to the water — that exposed 124 × 14 = 1736 mm² is the working area.

Push each plate in until its end sits 5 mm inside the bellmouth or rear block pocket. That pocket is the seal: it holds the end and keeps water from sneaking behind the plate.

5 Bellmouth and rear block

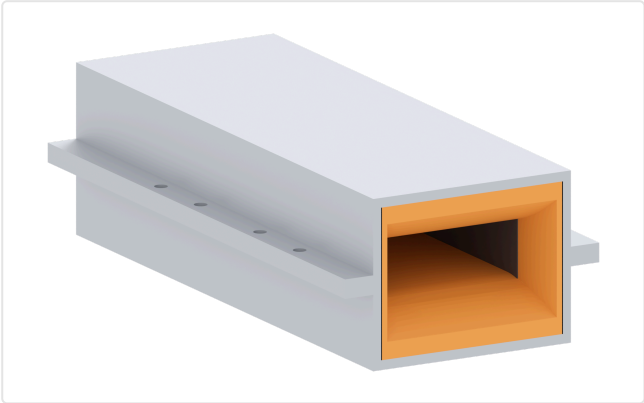
Four 3 mm pegs each. The bellmouth takes the 48 × 30 mouth down to the 40 × 14 bore over 20 mm — this is what keeps the inlet flow attached.

The rear block carries the **4.4 mm lead exit** at 13.5 mm off the mid-plane, one per half. Thread the magnet wires out through it and seal the hole with epoxy after. Leave the outlet straight: a diverging nozzle was measured to drop boat speed from 0.3 to 0.2 m/s through recirculation.

6 Close the clamshell

Lower tray assembly into the bottom half, upper into the top half, then bring the halves together. Drop 8 × M3 through the flanges and tighten in a cross pattern, a little at a time, so they seat flat instead of rocking.

The flanges are 8 mm strips meeting edge to edge around Ø3.4 holes, so location comes from the bolt shanks. If the pump will be knocked around, add two 3 mm dowels through the flanges before final torque.



Fit and clearance

Joint	Clearance	Consequence if wrong
magnet in tray cavity	0.3 per side	press fit cracks tiles
cap in rim ledge	0.15 per side	cap lifts, magnets float up into the steel
tray in steel cavity	0.1 per side	tray rattles, wears through the 1.5 mm floor
electrode in slot	0.2 total	water bypasses the plate, current skips the bore
electrode end in pocket	5 mm deep	short end = leak behind the electrode
flange faces	0 (metal to metal)	any gasket = air gap in the return path
M3 in flange hole	Ø3.4	standard M3 clearance

Before you power it

- ☐ Both halves flush at the split line, 8 bolts seated, no visible gap in the flange joint
- ☐ Compass: every row on a face points the same way, the two faces oppose
- ☐ Tray floor unbroken over the whole magnet area — hold it up to a light
- ☐ Electrodes fully home in their end pockets, not shorting to steel through a scratch
- ☐ Lead exit sealed; bore clear down its whole 124 mm — no epoxy drips
- ☐ Fuse in the supply line; runs under 2 minutes, ventilated (Cl_2)

> **Model status.** Dimensions are read back off the delivered solid, not copied from the spec. Magnets are modelled as four 120 × 9.7 × 10 bars per face rather than 96 loose tiles: same envelope, same 40 mm coverage, easier to read in a render.

Magnetide v21-r2e · generated from `Magnetide-v21r2e-assembly.step` · 18 solids · bore verified clear 40 × 14 × 124 · steel blank 107.83 × 150 × 2.5